Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum xp(x)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

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Classified Practice **Fundamentals of Random Variables**

Q001 **Written**

For the sum of two dice, construct the probability distribution and find P$5 <= X <= 8$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

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Q003 Written

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

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Q005 **Written**

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 **Written**

Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

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Q007 **Written**

For tails in four fair coins, find E(3X+2), E(-4X+1), and EX^2.

Q008 **Written**

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

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Q009 **Written**

Solve both source Exercise groups of transformed expectations for coins and for balls.

Q010 **Written**

For black balls drawn from the stated box, find EX and VX.

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Q011 **Written**

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

Q012 **Written**

Find EX and VX for all five source Exercise random variables.

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Q013 **Written**

Find the standard deviation of tails when two biased coins have P(head)=4/7 and P(tail)=3/7.

Q014 **Written**

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

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Q015 **Written**

Find the standard deviation for both source probability-table exercises.

Q016 **Written**

For the number of winning tickets in the source draw, find VY and sigma$Y$ for Y=2X+1 and Y=-3X-2.

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Q017 Written

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 **Written**

For a single die, find VY and sigma$Y$ for Y=X+2, Y=3X+1 and Y=-2X+4.

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Q019 **Written**

Solve the two source transformation problems: a net prize after an entry fee, and finding a,b from E(aX+b)=7/3 and sigma(aX+b)=3.

Q020 **Written**

Solve the two source Try problems involving net winnings and determining a,b for a die transformation.

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Q021 **Written**

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 MCQ

A probability distribution must have probabilities that sum to:

Choose the correct option.

A 0

B 1

C the number of outcomes

D the mean

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Q023 MCQ

For two fair dice, P$X = 2$ is:

Choose the correct option.

A 1/6
B 1/36
C 2/36
D 0

Q024 **MCQ**

The number of heads in three coins can be:

Choose the correct option.

A 1,2,3

B 0,1,2,3

C 0,3 only

D any integer

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Q025 *MCQ*

The expected value is calculated as:

Choose the correct option.

A sum P$X = x$
B sum xP$X = x$
C largest x
D variance squared

Q026 MCQ

The mean number of tails in four fair coins is:

Choose the correct option.

A 1

B 2

C 3

D 4

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Q027 MCQ

For a fair die, EX is:

Choose the correct option.

A 3

B 3.5

C 4

D 6

Q028 **MCQ**

E(aX+b) equals:

Choose the correct option.

A aEX

B aEX+b

C EX+b only

D a+b

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Q029 MCQ

In general, $\mathrm{E}X^2$ is:

Choose the correct option.

A $E[X]^2$

B equal to $\mathrm{E}X$

C a weighted square sum

D always zero

Q030 MCQ

If EX=2, then E(3X+2) is:

Choose the correct option.

A 6

B 8

C 10

D 12

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Q031 MCQ

The variance identity is:

Choose the correct option.

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D sqrt V

Q032 MCQ

Variance can never be:

Choose the correct option.

A positive

B zero

C negative

D fractional

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Q033 MCQ

The first step in a variance problem is to:

Choose the correct option.

A draw the distribution
B take a square root
C ignore probabilities
D add the values

Q034 MCQ

Standard deviation is:

Choose the correct option.

A VX

B sqrt$V(X)$

C VX squared

D EX

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Q035 **MCQ**

Standard deviation has units:

Choose the correct option.

A squared units
B the same units as X
C no units
D units cubed

Q036 MCQ

V(aX+b) equals:

Choose the correct option.

A aVX+b

B a squared times VX

C |a|VX

D VX+b

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Q037 **MCQ**

sigma(aX+b) equals:

Choose the correct option.

A a sigma$X$

B |a| sigma$X$

C a squared sigma$X$

D sigma$X$+b

Q038 **MCQ**

For Y=X+3, the variance is:

Choose the correct option.

A VX+3

B VX

C 3VX

D 9VX

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Q039 **MCQ**

A net prize after a fixed fee is modeled as:

Choose the correct option.

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ

If a>0 and sigma(aX+b)=3, then a is:

Choose the correct option.

A 3 sigma$X$

B 3/sigma$X$

C sigma$X/3$

D 3+b

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Q041 MCQ

Changing b in aX+b changes:

Choose the correct option.

A variance only

B standard deviation only

C the mean only

D nothing

Q042 MCQ

If EX=2 and EX^2=7, then VX is:

Choose the correct option.

A 3

B 5

C 7

D 9

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Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 **Concept Quiz · Written**

For Y=4X-1 with VX=2, find VY and sigma$Y$.

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Quiz WRITTEN 3 **Concept Quiz · Written**

For a fair die, find E(2X+5).

Quiz WRITTEN 4 **Concept Quiz · Written**

If sigma$X$=2 and EX=3, find sigma(-5X+7).

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Quiz MCQ 5 **Concept Quiz · MCQ**

If VX=5, V(3X+1) is:

Choose the correct option.

A 5

B 8

C 15

D 45

Quiz MCQ 6 **Concept Quiz · MCQ**

For a fair die EX is:

Choose the correct option.

A 2.5

B 3

C 3.5

D 4

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Quiz MCQ 7 Concept Quiz · MCQ

A probability table total must be:

Choose the correct option.

A 0

B 1

C 2

D the mean

Quiz MCQ 8 **Concept Quiz · MCQ**

If sigma$X$=4, sigma(X-9) is:

Choose the correct option.

A 0
B 4
C 13
D 16

Answer key

Use the question number shown on each page.

Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 $<= X <=$ 8=20/36=5/9.

Q002 Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Q005 Use EX=sum xP$X = x$ after constructing each distribution.

Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Q007 Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 Apply linearity for aX+b and use the weighted square sum for X^2.

Q009 For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 $EX = 6/7; VX = 20/49$.

Q011 Construct each probability table, then use VX=EX^2-EX^2.

Q012 Use the distribution table for each experiment and the variance identity.

Q013 The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

Q014 Use sigma$X$=sqrt$V(X)$ after evaluating the variance from the table.

Q015 Compute EX, EX^2, then sigma$X$=sqrt$E(X^2) - E(X)^2$.

Q016 Use $V(aX + b) = a^2VX$ and sigma(aX+b)=|a|sigma$X$.

Q017 Apply the squared multiplier to VX and the absolute multiplier to sigma$X$.

Q018 Use the variance and standard-deviation transformation formulas for each Y.

Q019 Use $E(aX + b) = aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 B

Q023 B

Q024 B

Q025 B

Q026 B

Q027 B

Q028 B

Q029 C

Q030 C

Q031 B

Q032 C

Q033 A

Q034 B

Q035 B

Q036 B

Q037 B

Q038 B

Q039 B

Q040 B

Q041 C

Q042 A

Quiz WRITTEN 1 EX=3/2 and VX=3/4

Quiz WRITTEN 2 VY=32 and sigma$Y$=4 sqrt2

Quiz WRITTEN 3 12

Quiz WRITTEN 4 10

Quiz MCQ 5 D

Quiz MCQ 6 C

Quiz MCQ 7 B

Quiz MCQ 8 B